

# SQL Customer Analysis Project

E-commerce Sales & Customer Insights Analysis

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Prepared By:  
Ankita Raj



# Project Objective

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The objective of this project is to analyze e-commerce customer transaction data using SQL and generate business insights related to sales trends, customer behavior, repeat purchases, and product performance.

## THE PROJECT FOCUSES ON:



**Monthly Sales  
Analysis**



**Top-Selling  
Products**



**Repeat Customer  
Identification**



**Customer  
Retention**



**Average Order  
Value**

# Dataset Overview

90

TOTAL ORDERS

10

TOTAL CUSTOMERS

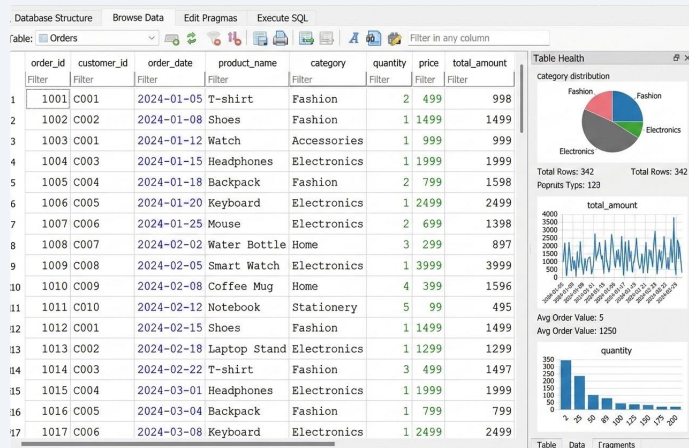
JAN 2024 – DEC 2024

TIME PERIOD

## Product Categories

Fashion, Electronics, Home, Stationery

The dataset contains customer order transactions including product details, quantity, sales value, and purchase dates.



# Tools & Technologies

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## **SQLite**

SQL Query Execution

## **DB Browser**

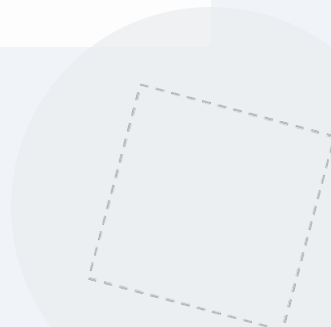
Database Management

## **Excel**

Charts & Visualization

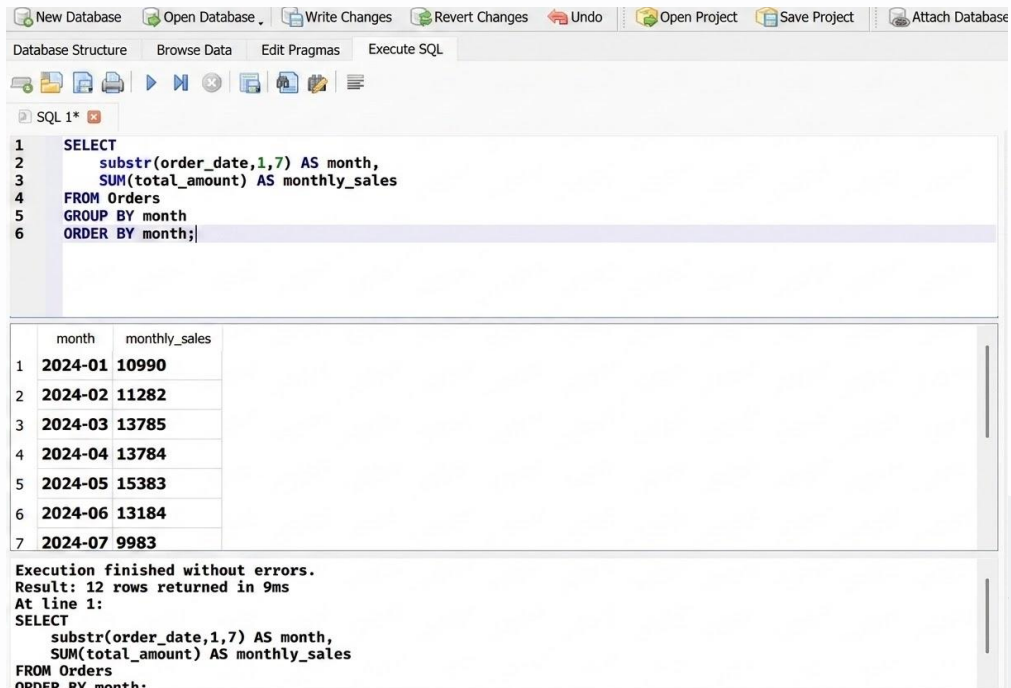
## **PowerPoint / Canva**

Presentation & Reporting



# Monthly Sales Trend Analysis

- Sales showed steady growth during the first half of the year.
- Peak sales were observed during May and June.
- Lower sales were observed in July and August.
- Seasonal buying behavior can be identified from monthly variations.



The screenshot displays a database management interface with a menu bar (New Database, Open Database, Write Changes, Revert Changes, Undo, Open Project, Save Project, Attach Database) and a toolbar. The 'Execute SQL' tab is active, showing a SQL query in a text editor. Below the editor, a table displays the results of the query. The table has two columns: 'month' and 'monthly\_sales'. The results show a steady increase in sales from January to June, with a slight dip in July.

```
1 SELECT
2     substr(order_date,1,7) AS month,
3     SUM(total_amount) AS monthly_sales
4 FROM Orders
5 GROUP BY month
6 ORDER BY month;
```

|   | month   | monthly_sales |
|---|---------|---------------|
| 1 | 2024-01 | 10990         |
| 2 | 2024-02 | 11282         |
| 3 | 2024-03 | 13785         |
| 4 | 2024-04 | 13784         |
| 5 | 2024-05 | 15383         |
| 6 | 2024-06 | 13184         |
| 7 | 2024-07 | 9983          |

Execution finished without errors.  
Result: 12 rows returned in 9ms  
At line 1:  
SELECT  
 substr(order\_date,1,7) AS month,  
 SUM(total\_amount) AS monthly\_sales  
FROM Orders  
ORDER BY month;

# Top-Selling Product Analysis

- **T-shirts and Water Bottles** recorded high sales volume.
- **Electronics products** generated high revenue despite lower quantity.
- **Smart Watches** contributed significantly to revenue generation.

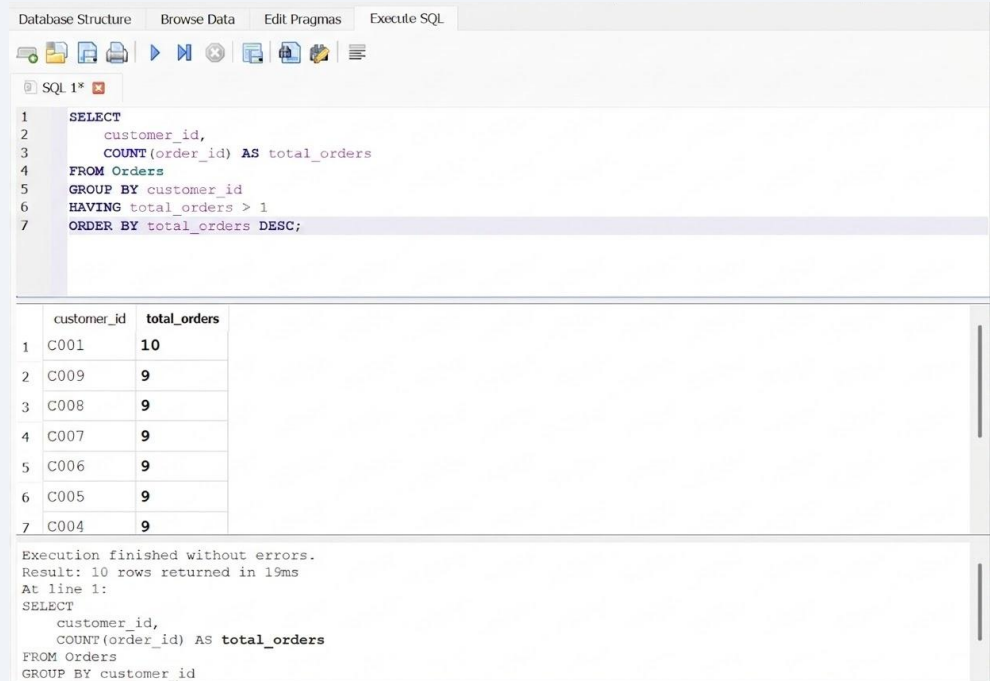
```
1 SELECT
2     product_name,
3     SUM(quantity) AS total_quantity_sold
4 FROM Orders
5 GROUP BY product_name
6 ORDER BY total_quantity_sold DESC;
```

|   | product_name | total_quantity_sold |
|---|--------------|---------------------|
| 1 | Notebook     | 38                  |
| 2 | Water Bottle | 29                  |
| 3 | T-shirt      | 27                  |
| 4 | Coffee Mug   | 25                  |
| 5 | Backpack     | 14                  |
| 6 | Shoes        | 13                  |
| 7 | Mouse        | 12                  |

```
Execution finished without errors.
Result: 12 rows returned in 44ms
At line 1:
SELECT
    product_name,
    SUM(quantity) AS total_quantity_sold
FROM Orders
GROUP BY product_name
```

# Repeat Customer Analysis

- Multiple customers placed repeated orders during the year.
- Repeat customers indicate customer satisfaction and loyalty.
- Retention-focused marketing strategies can increase profitability.



The screenshot shows a database management interface with tabs for 'Database Structure', 'Browse Data', 'Edit Pragmas', and 'Execute SQL'. The 'Execute SQL' tab is active, displaying a SQL query in a text editor. Below the editor, a table shows the results of the query, listing customer IDs and their total orders. At the bottom, a status message indicates the execution was successful and returned 10 rows.

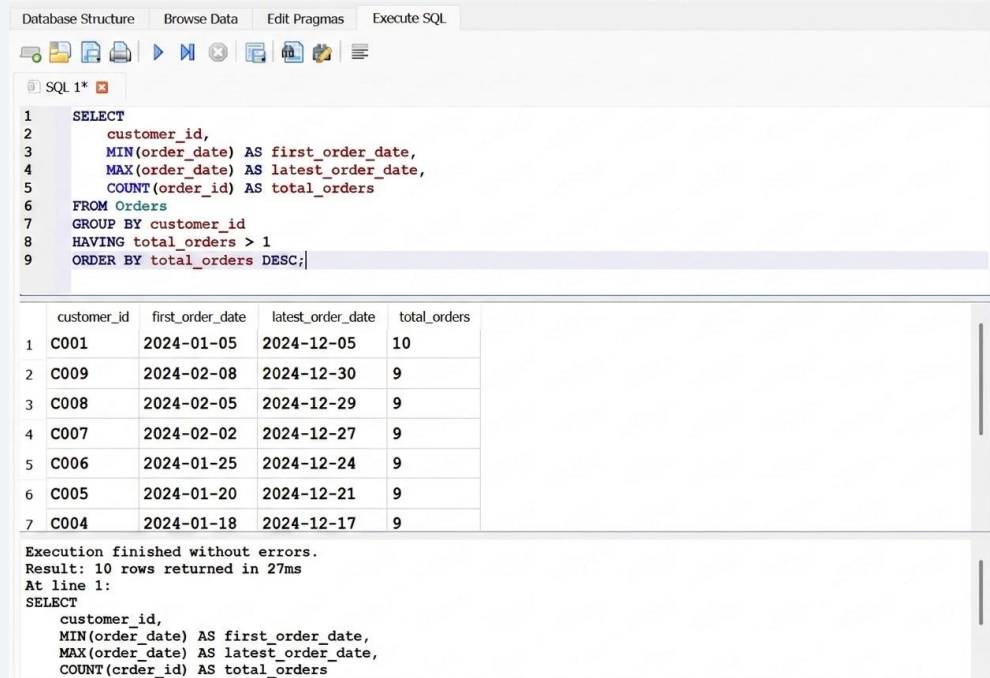
```
1 SELECT
2   customer_id,
3   COUNT(order_id) AS total_orders
4 FROM Orders
5 GROUP BY customer_id
6 HAVING total_orders > 1
7 ORDER BY total_orders DESC;
```

|   | customer_id | total_orders |
|---|-------------|--------------|
| 1 | C001        | 10           |
| 2 | C009        | 9            |
| 3 | C008        | 9            |
| 4 | C007        | 9            |
| 5 | C006        | 9            |
| 6 | C005        | 9            |
| 7 | C004        | 9            |

Execution finished without errors.  
Result: 10 rows returned in 19ms  
At line 1:  
SELECT  
 customer\_id,  
 COUNT(order\_id) AS total\_orders  
FROM Orders  
GROUP BY customer\_id

# Customer Retention Analysis

- Several customers continued purchasing throughout the year.
- Returning customers contribute to **stable business revenue**.
- Retention strategies are **more cost-effective** than acquiring new customers.



The screenshot shows a database management interface with tabs for 'Database Structure', 'Browse Data', 'Edit Pragmas', and 'Execute SQL'. The 'Execute SQL' tab is active, displaying an SQL query and its results. The query is a SELECT statement that groups data by customer\_id, showing the first and latest order dates and the total number of orders. The results are displayed in a table with 4 columns: customer\_id, first\_order\_date, latest\_order\_date, and total\_orders. The table contains 7 rows of data, sorted by total\_orders in descending order. Below the table, a status message indicates that the execution finished without errors and that 10 rows were returned in 27ms. The SQL query is as follows:

```
1 SELECT
2     customer_id,
3     MIN(order_date) AS first_order_date,
4     MAX(order_date) AS latest_order_date,
5     COUNT(order_id) AS total_orders
6 FROM Orders
7 GROUP BY customer_id
8 HAVING total_orders > 1
9 ORDER BY total_orders DESC;
```

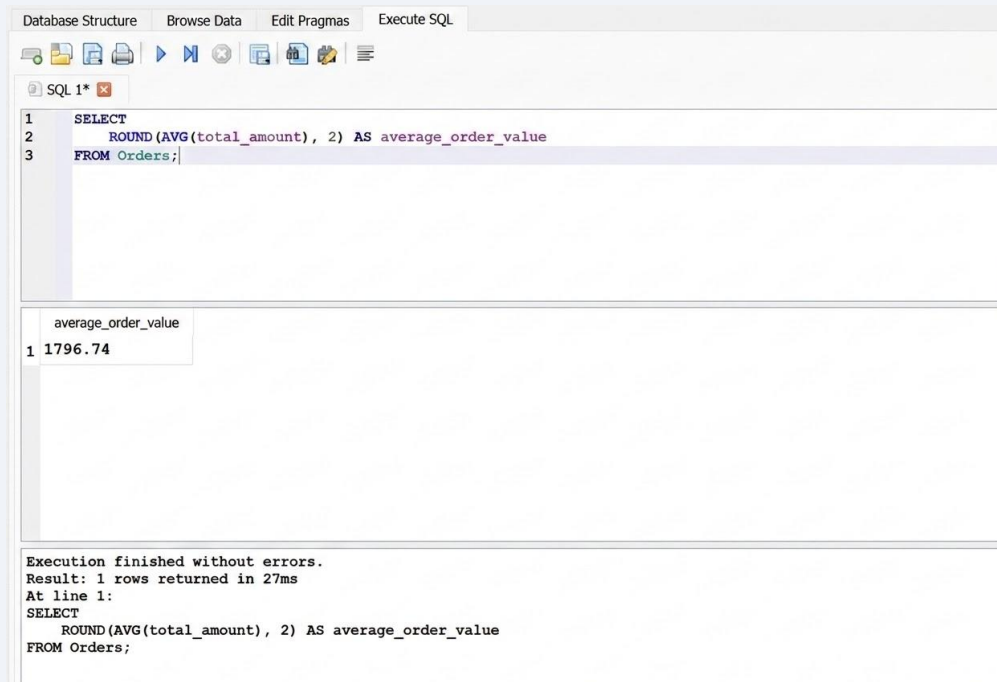
|   | customer_id | first_order_date | latest_order_date | total_orders |
|---|-------------|------------------|-------------------|--------------|
| 1 | C001        | 2024-01-05       | 2024-12-05        | 10           |
| 2 | C009        | 2024-02-08       | 2024-12-30        | 9            |
| 3 | C008        | 2024-02-05       | 2024-12-29        | 9            |
| 4 | C007        | 2024-02-02       | 2024-12-27        | 9            |
| 5 | C006        | 2024-01-25       | 2024-12-24        | 9            |
| 6 | C005        | 2024-01-20       | 2024-12-21        | 9            |
| 7 | C004        | 2024-01-18       | 2024-12-17        | 9            |

Execution finished without errors.  
Result: 10 rows returned in 27ms  
At line 1:  
SELECT  
customer\_id,  
MIN(order\_date) AS first\_order\_date,  
MAX(order\_date) AS latest\_order\_date,  
COUNT(crder\_id) AS total\_orders



# Average Order Value (AOV)

- Average Order Value measures the average spending per customer transaction.
- Higher AOV indicates stronger revenue generation per order.



The screenshot displays a database management tool interface with tabs for Database Structure, Browse Data, Edit Pragmas, and Execute SQL. The Execute SQL tab is active, showing a query in a text editor:

```
1 SELECT
2   ROUND (AVG(total_amount), 2) AS average_order_value
3 FROM Orders;
```

Below the query editor, the results are displayed in a table with one column, `average_order_value`, and one row containing the value `1796.74`.

At the bottom, a status message indicates: "Execution finished without errors. Result: 1 rows returned in 27ms. At line 1: SELECT ROUND (AVG(total\_amount), 2) AS average\_order\_value FROM Orders;"

# Key Business Insights



## Market Leaders

Fashion and Electronics emerged as the highest-performing categories across the portfolio.

## Premium Impact

High-value electronic products generated a strong revenue contribution.

## Consumer Trends

Analysis of monthly sales revealed significant seasonal demand fluctuations throughout the year.

## Loyalty Patterns

Certain key customer segments demonstrated clear repeat purchasing behavior.



## Retention Success

Customer retention positively impacted overall sales performance, stabilizing long-term growth.

# Business Recommendations



## Customer Loyalty

Introduce loyalty programs for repeat customers to drive long-term engagement.

## Strategic Retention

Improve retention through personalized offers and discounts based on behavior.

## Sales & Marketing

Increase marketing activity during low-sales months to balance annual revenue.

## Targeted Campaigns

Promote high-performing products through campaigns focused on key segments.

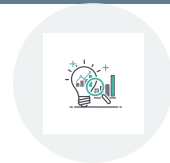


## Value Growth

Bundle complementary products to increase average order value across transactions.

# SQL Skills Demonstrated

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## Technical Proficiency

- SELECT Statement
- GROUP BY & ORDER BY Clauses
- HAVING Clause
- Aggregate Functions (SUM, AVG, COUNT)

## Analytical Expertise

- Sales Data & Retention Analysis
- Customer Segmentation
- Data-Driven Decision Making

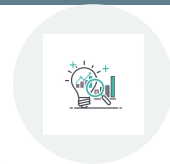


## Tools Used

- SQLite & DB Browser
- Microsoft Excel
- Microsoft PowerPoint

# Conclusion

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## Project Success

This project successfully demonstrated the use of SQL for business data analysis.



## Key Insights

SQL queries identified vital insights in customer behavior, sales trends, and product performance.



## Data-Driven Impact

The project highlights how SQL supports efficient data-driven decision-making in e-commerce.